Plasma-Echo-Constrained Kinetic Memory

A phase-coherent intervention and identifiability framework for reduced kinetic models

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Abstract. Plasma echoes expose a central distinction in kinetic theory: decay of a resolved moment under phase mixing does not imply that the underlying phase information has been destroyed. This paper develops an echo-centred intervention framework for probing that distinction in reduced kinetic descriptions. Starting from a two-pulse phase ansatz, we derive the difference-mode wavenumber and rephasing time, then generalise the Gaussian free-streaming coherence to anisotropic velocity distributions and non-collinear wavevectors. A vector projection formula identifies the post-pulse time of maximal coherence and quantifies the residual mismatch when complete rephasing is geometrically impossible. We introduce a factorial pulse-interaction contrast that subtracts single-pulse responses, derive the cubic attenuation factor produced by integrated Brownian velocity diffusion over a finite preconditioning interval, and formulate a two-term attenuation family whose parameters can be distinguished only under an identifiable pulse-separation design. The exact Mori–Zwanzig projection identity is used to clarify what an echo intervention can and cannot reveal about unresolved memory. Numerical work is deliberately bounded: a deterministic synthetic scalar signal is evaluated over 30 fixed-seed chronological holdouts, where 24-sample history windows reduce one-step RMSE by 26.83% on average relative to a one-sample baseline. This is a synthetic forecasting result, not evidence of a learned physical memory kernel. The manuscript provides a falsifiable protocol for a future converged Vlasov–Poisson study; it does not claim a self-consistent kinetic simulation, experimental validation, or a new plasma mechanism.

Keywords: plasma echo; phase mixing; Landau damping; velocity-space diffusion; Mori–Zwanzig; non-Markovian closure; intervention contrast; identifiability; reduced-order modelling.

1 Research posture and contribution

The question addressed here is not whether plasma echoes exist; that phenomenon has been observed experimentally and is embedded in the modern theory of nonlinear kinetic dynamics (Malmberg et al., 1968; Su and Oberman, 1968; Ichikawa and Suzuki, 1969; Grenier, Nguyen, and Rodnianski, 2022). The question is methodological: can a deliberately timed second perturbation act as an intervention that discriminates between (i) apparent decay caused by reversible phase cancellation in unresolved velocity space and (ii) a specified model of irreversible or effectively irreversible attenuation? A useful answer must separate kinematic prediction, model-dependent amplitude response, and inference from data.

The central object is an echo-mode response measured in a protocol with four arms: neither pulse, first pulse only, second pulse only, and both pulses. The interaction contrast removes the single-pulse contributions and isolates the lowest-order cross-pulse component. Under a narrow perturbative regime, this contrast scales bilinearly with the two pulse amplitudes. Its time dependence is then compared with a kinematic coherence envelope and with alternative attenuation families. This strategy does not make the echo a direct measurement of the Mori–Zwanzig kernel. Instead, it creates a controlled, phase-sensitive observable with which candidate reduced closures can be challenged.

The paper makes four bounded contributions. First, it gives a compact derivation of the one-dimensional rephasing law and a covariance-weighted vector generalisation. Second, it derives the Gaussian coherence magnitude from the characteristic function of the velocity distribution and distinguishes full vector rephasing from a post-pulse constrained maximum. Third, it relates a cubic-in-separation attenuation factor to integrated Brownian velocity diffusion, while showing why a second time dependence should be retained as a competing phenomenological term rather than silently absorbed into a single coefficient. Fourth, it specifies an identifiable pulse-separation design and a falsification protocol connecting echo-centred interventions to reduced-memory modelling. The synthetic regression experiment is supporting software verification only; it is not a kinetic experiment.

1.1 Claim boundary

A formula derived from a chosen phase ansatz is exact within that ansatz, but the ansatz is not the nonlinear Vlasov–Poisson system. The attenuation family is an interpretable reduced model; its coefficients are not asserted to equal a measured Coulomb collision frequency. The delay-window result concerns one constructed time series. No field solver, particle-in-cell data, laboratory measurement, MMS observation, or trained neural operator is used in this release. The distinction between mathematical consequence and physical validation is maintained throughout.

2 Kinetic context and closest prior work

2.1 Echoes, damping, and nonlinear interactions

Landau damping concerns the self-consistent evolution of fields coupled to a continuum of particle velocities. Phase mixing transfers information into progressively finer velocity dependence, so a low-order density or field moment may become small even while the fine-grained distribution retains structure. The plasma-wave echo experiment of Malmberg, Wharton, Gould, and O'Neil established that a later perturbation can recover a macroscopic signal from such phase-mixed structure (Malmberg et al., 1968). Su and Oberman analysed

collisional damping of plasma echoes, including the sensitivity of attenuation to weak collisions (Su and Oberman, 1968). Ichikawa and Suzuki subsequently compared relaxation and Fokker–Planck-type collision models and reported that the latter can contain both cubic and linear dependence on the pulse separation in the relevant damping expression (Ichikawa and Suzuki, 1969). This is a direct warning against treating one cubic exponential as a universal collisional law.

At the nonlinear mathematical level, echo interactions are not confined to a single two-pulse event. Grenier, Nguyen, and Rodnianski constructed solutions near stable Penrose data in which high-frequency waves interact in a cascade of echoes while large-time damping can still occur (Grenier, Nguyen, and Rodnianski, 2022). Their theorem concerns the Vlasov–Poisson dynamics and establishes a much richer regime than the reduced envelope used here. The present manuscript does not approximate or reproduce that theorem. Its two-pulse geometry is an analytic probe and a proposed diagnostic for future kinetic computation.

2.2 Projection memory and learned closures

The Mori–Zwanzig programme formalises the influence of unresolved variables on resolved observables through an instantaneous term, a history integral, and a term depending on unresolved initial conditions. The foundational formulations of Zwanzig and Mori, and the later optimal-prediction treatment of Chorin, Hald, and Kupferman, make clear that memory arises naturally when a high-dimensional dynamical system is projected onto a smaller set of variables (Zwanzig, 1961; Mori, 1965; Chorin, Hald, and Kupferman, 2000). The memory kernel is generally not observed directly: it is an object in an exact identity whose practical reconstruction depends on projection choice, data, and closure assumptions.

A particularly close modern precedent is the 2026 neural-operator closure for Landau damping by Burles, Camporeale, and Pezzi. Their learned non-Markovian closure uses a trailing history window and is trained within a differentiable fluid solver to reproduce kinetic effects in one-dimensional electrostatic settings (Burles, Camporeale, and Pezzi, 2026). Accordingly, finite-memory plasma closures and neural operators are not claimed as novel here. The narrower objective is an echo-conditioned intervention and identifiability analysis that can serve as a test protocol for reduced models, including but not limited to learned closures.

3 Mathematical setup and notation

Consider a one-dimensional spatial coordinate x, velocity v, and a perturbation $f_1(x, v, t)$ about a spatially homogeneous equilibrium $f_0(v)$. The linearised electrostatic Vlasov equation is

$$\partial_t f_1 + v\,\partial_x f_1 + \frac{q}{m} E_1\,\partial_v f_0 = 0, \tag{1}$$

with the field coupled to the perturbed charge density through Poisson's equation. Equation (1) is stated to locate the kinetic setting; it is not solved numerically in this work. In the freely streaming part, a spatial Fourier component with wavenumber k accumulates the velocity-dependent phase $\exp(-\mathrm{i}kvt)$. Integrating over v can suppress a low-order moment through destructive interference.

Let pulse 1 act at $t = 0$ with wavenumber k_1 and dimensionless amplitude a_1. Let pulse 2 act at $t = \tau$ with wavenumber k_2 and amplitude a_2. We adopt the sign convention in which the mixed mode has $k_3 = k_2 - k_1$. The observable is a complex Fourier moment $M_{k_3}(t)$ or a calibrated field proxy at that mode. The pulse amplitudes are control parameters, not measured field amplitudes. The normalised coherence derived below is the velocity-integrated phase factor; it is not by itself a prediction of the absolute electric-field amplitude.

3.1 Epistemological classification

The P1–P5 labels separate the logical status of claims. They prevent a reduced calculation from being silently elevated into a physical result. Table 3 records the principal classes used below.

4 Two-pulse phase geometry

4.1 Scalar rephasing

In the selected perturbative phase ansatz, the mixed contribution after the second pulse contains a velocity phase proportional to

$$\phi(v, t) = v\,(k_2\tau - k_3 t), \qquad k_3 = k_2 - k_1. \tag{2}$$

The phase becomes independent of v when its coefficient vanishes. Thus

$$t_e = \frac{k_2\tau}{k_2 - k_1} = \frac{k_2}{k_3}\tau, \qquad k_2 > k_1 > 0. \tag{3}$$

Equation (3) is a P1 kinematic identity conditional on Eq. (2). It does not determine the field response, the amplitude of the echo, or the stability of a nonlinear kinetic solution. It also makes the post-second-pulse condition explicit: $t_e > \tau$ for positive k_1 and $k_2 > k_1$.

4.2 Gaussian velocity coherence

Let $p(v)$ be a normalised one-dimensional Gaussian velocity distribution with mean u and variance σ_v^2. The velocity-integrated phase factor is its characteristic function,

$$C(t) = \int_{-\infty}^{\infty} p(v)\exp[-\mathrm{i}v(k_2\tau - k_3 t)]\,\mathrm{d}v \tag{4}$$

$$= \exp[-\mathrm{i}uq(t)]\exp[-\tfrac{1}{2}\sigma_v^2 q(t)^2], \tag{5}$$

where $q(t) = k_2\tau - k_3 t$. The magnitude is

$$|C(t)| = \exp\left[-\frac{\sigma_v^2}{2}(k_2\tau - k_3 t)^2\right]. \tag{6}$$

The magnitude has a unit peak at t_e and a temporal standard deviation $1/(\sigma_v|k_3|)$ in the Gaussian envelope. Its full width at half maximum is

$$\mathrm{FWHM}_t = \frac{2\sqrt{2\ln 2}}{\sigma_v|k_3|}. \tag{7}$$

A narrower velocity distribution therefore produces a broader time-domain rephasing envelope, whereas a larger $|k_3|$ sharpens it. The mean velocity u changes the complex phase but not the magnitude in Eq. (6). The profile is a normalised free-streaming moment factor, not a self-consistent field amplitude.

Table 1: Research position relative to established results. The final column states the boundary respected in this manuscript.

Work	Established contribution	Relevance to the present problem	Claim boundary here
Malmberg et al. (1968)	Experimental observation of a plasma-wave echo.	Establishes that a later perturbation can recover a macroscopic signal from phase-mixed structure.	Echo physics is prior art, not a new discovery.
Su–Oberman (1968); Ichikawa–Suzuki (1969)	Collisional attenuation of temporal and spatial echoes under simplified collision models.	Shows that the attenuation law depends on the collision model and pulse interval.	The reduced coefficients here are not physical collision frequencies.
Grenier et al. (2022)	Nonlinear echo cascades near stable Penrose data for Vlasov–Poisson.	Establishes that echo interactions can occur in a mathematically complex nonlinear setting.	No kinetic theorem or echo cascade is reproduced here.
Zwanzig (1961); Mori (1965)	Projection-based non-Markovian dynamics and memory functions.	Provides the exact structural language for unresolved degrees of freedom.	No exact memory kernel is reconstructed.
Chorin et al. (2000)	Optimal prediction connected to Mori–Zwanzig representation.	Motivates history-aware reduced prediction and the role of conditional information.	The scalar regression is only a synthetic baseline.
Burles et al. (2026)	Neural-operator, finite-history closure for Landau damping.	Direct adjacent precedent for learned non-Markovian plasma closure.	Finite-memory closure is expressly not claimed as novel.

Table 2: Notation and dimensions. Dimensionless quantities may be obtained by normalising length by λ_D and time by ω_p^{-1}.

Symbol	Meaning	Dimension
$f_0(v)$	Equilibrium velocity density	model dependent
k_1, k_2, k_3	Pulse and mixed-mode wavevectors	L^{-1}
τ, t, t_e	Pulse gap, observation and echo time	T
a_1, a_2	Small pulse amplitudes	dimensionless
$\bar{\boldsymbol{v}}, \Sigma_v$	Mean and covariance of velocity	LT^{-1}; L^2T^{-2}
D_v	Velocity-space diffusion coefficient	L^2T^{-3}
β	Effective linear-time attenuation rate	T^{-1}
P, Q	Resolved projection and complement	operators
$\mathcal{K}(t)$	Mori–Zwanzig memory kernel	operator valued
$\Delta_{12}M$	Factorial pulse-interaction contrast	observable

4.3 Vector extension and incomplete rephasing

For a d-dimensional velocity vector with Gaussian distribution $\boldsymbol{v} \sim \mathcal{N}(\bar{\boldsymbol{v}}, \Sigma_v)$, where Σ_v is positive definite, define the mismatch vector

$$\boldsymbol{q}(t) = \boldsymbol{k}_2\tau - \boldsymbol{k}_3 t. \tag{8}$$

The corresponding characteristic function is

$$C_d(t) = \exp[-\mathrm{i}\bar{\boldsymbol{v}}^\top \boldsymbol{q}(t)] \exp\left[-\tfrac{1}{2}\boldsymbol{q}(t)^\top \Sigma_v \boldsymbol{q}(t)\right]. \tag{9}$$

The magnitude is maximised by minimising the quadratic mismatch. Differentiating $J(t) = \boldsymbol{q}(t)^\top \Sigma_v \boldsymbol{q}(t)$ gives the unconstrained optimum

$$t_*^{\mathrm{unc}} = \tau \, \frac{\boldsymbol{k}_3^\top \Sigma_v \boldsymbol{k}_2}{\boldsymbol{k}_3^\top \Sigma_v \boldsymbol{k}_3}. \tag{10}$$

Since an echo is observed after pulse 2, the physically admissible reduced optimum is $t_* = \max(\tau, t_*^{\mathrm{unc}})$. The residual coherence is then determined by the mismatch at that constrained time.

Proposition 1 (criterion for complete vector rephasing). Under a positive-definite covariance Σ_v, complete rephasing at a post-pulse time $t > \tau$ occurs if and only if $\boldsymbol{k}_2 = c\boldsymbol{k}_3$ for some scalar $c > 1$. In that case Eq. (10) gives $t_e = c\tau$ and zero mismatch. Otherwise the admissible post-pulse maximum is strictly below one.

Table 3: Epistemic status of the main statements.

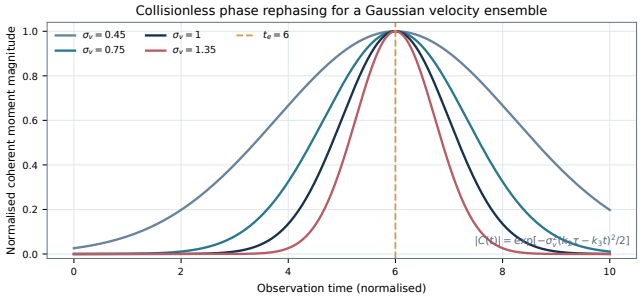
Class	Statement	Status in this paper
P1	Gaussian characteristic function; scalar and vector rephasing relations.	Exact under the stated Gaussian/free-streaming phase assumptions.
P2	A physical variational or dynamical selection principle.	No such claim is made.
P3	Universal scaling exponent or universality class.	No such claim is made; no scaling study is performed.
P4	Echo-conditioned probing as a structural bridge to projected memory dynamics.	Formal correspondence and proposed protocol; physical identification remains open.
P5	Flow diagrams and conceptual information pathways.	Explanatory schematics only, not empirical causal graphs.

Proof. Complete rephasing requires $\boldsymbol{q}(t) = \boldsymbol{0}$, or $\boldsymbol{k}_2\tau = \boldsymbol{k}_3 t$. For $\tau > 0$ and $t > \tau$, this is equivalent to $\boldsymbol{k}_2 = c\boldsymbol{k}_3$ with $c = t/\tau > 1$. If the vectors are not collinear, or if their positive ratio is not greater than one, no admissible post-pulse time can make the vector mismatch zero. Because Σ_v is positive definite, $\boldsymbol{q}^\top \Sigma_v \boldsymbol{q} > 0$ for every nonzero $\boldsymbol{q}$, so the coherence magnitude in Eq. (9) is strictly below one. The scalar minimiser follows from the positive quadratic form and Eq. (10). □

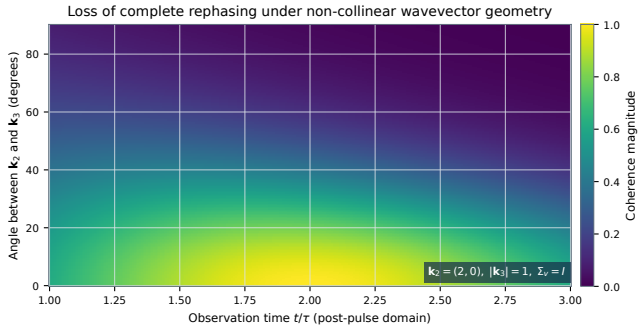
This proposition is a geometry statement about a Gaussian phase ensemble. It is not a theorem about the full nonlinear Vlasov–Poisson system. It shows why a one-dimensional formula should not be applied unchanged to oblique wavevector configurations. Figure 1b displays the coherence loss in a normalised two-dimensional geometry.

5 Intervention contrast and perturbative response

A raw echo-mode trace may contain background oscillations, single-pulse transients, and direct responses to either perturbation. To isolate the cross-pulse contribution, measure four otherwise matched trajectories: M_{00} with neither pulse; M_{10} with pulse 1 only; M_{01} with pulse 2 only; and M_{11} with both



(a) Gaussian phase coherence for four velocity spreads. All profiles share the kinematic centre $t_e = 6$; the width is set by $\sigma_v|k_3|$.



(b) Normalised vector coherence in a post-pulse domain. Non-collinearity prevents complete cancellation of the velocity-dependent phase.

Figure 1: Phase-coherence geometry. These are analytic reduced-model curves, not measured plasma fields.

pulses. Define the factorial interaction contrast

$$\Delta_{12}M(t) = M_{11}(t) - M_{10}(t) - M_{01}(t) + M_{00}(t). \tag{11}$$

If the observable is sufficiently smooth in the pulse amplitudes, a Taylor expansion around $(a_1, a_2) = (0, 0)$ gives

$$\Delta_{12}M(t) = a_1 a_2\, \chi_{12}(t) + O(a_1^2 a_2 + a_1 a_2^2), \tag{12}$$

where

$$\chi_{12}(t) = \left.\frac{\partial^2 M(t; a_1, a_2)}{\partial a_1 \partial a_2}\right|_{a_1=a_2=0} \tag{13}$$

is the cross-pulse susceptibility. This is the response coefficient of interest, not the total response in M_{11}.

A practical finite-amplitude estimator is $\widehat{\chi}_{12} = \Delta_{12}M/(a_1 a_2)$. It should be evaluated over an amplitude ladder, with the estimate extrapolated toward the small-amplitude regime. A plateau in $\widehat{\chi}_{12}$ as both amplitudes decrease is a necessary diagnostic for bilinear response; it is not sufficient to prove that all higher-order terms are negligible. The order of the pulses, the background equilibrium, numerical resolution, and the observable definition must be held fixed across the four protocol arms. Figure 2a gives the causal and control-arm bookkeeping.

6 Velocity diffusion and attenuation structure

6.1 Integrated Brownian phase

A cubic dependence on the pulse separation can be derived for a simple stochastic velocity-diffusion model. Let the velocity during a preconditioning interval obey

$$\mathrm{d}v(s) = \sqrt{2D_v}\, \mathrm{d}W_s, \tag{14}$$

where W_s is standard Brownian motion and D_v has dimensions L^2T^{-3}. The stochastic part of the displacement accumulated over $[0, \tau]$ is

$$X_\tau = \sqrt{2D_v} \int_0^\tau W_s\, \mathrm{d}s. \tag{15}$$

Using $\mathrm{Cov}(W_s, W_r) = \min(s, r)$,

$$\mathrm{Var}(X_\tau) = 2D_v \int_0^\tau \int_0^\tau \min(s, r)\, \mathrm{d}s\, \mathrm{d}r \tag{16}$$

$$= 2D_v \int_0^\tau \left[2\int_0^s r\, \mathrm{d}r\right] \mathrm{d}s = \frac{2D_v\tau^3}{3}. \tag{17}$$

Because X_τ is Gaussian, the characteristic function of the spatial Fourier phase is

$$\mathbb{E}\left[\mathrm{e}^{-\mathrm{i}k_1 X_\tau}\right] = \exp\left[-\frac{k_1^2}{2}\mathrm{Var}(X_\tau)\right] = \exp\left[-\frac{D_v k_1^2 \tau^3}{3}\right]. \tag{18}$$

Thus the cubic factor follows exactly for integrated Brownian velocity over the specified preconditioning interval. This derivation is a stochastic reduced model, not a derivation of the complete two-pulse echo amplitude for a physical Coulomb collision operator. In particular, it does not include the post-second-pulse diffusion history, field feedback, nonlinear trapping, or the full kinetic collision integral.

6.2 A competing linear-time component

The literature on collisional echo damping demonstrates dependence on the collision model. Ichikawa and Suzuki discuss a Fokker–Planck-type result containing a term proportional to τ^3 and an additional contribution proportional to τ in the damping exponent (Ichikawa and Suzuki, 1969); Su and Oberman also analyse collisional damping under simplified assumptions (Su and Oberman, 1968). To avoid overstating the cubic-only expression, we use the two-term reduced family

$$A_{\mathrm{att}}(\tau; k_1) = \exp\left[-\alpha_3 k_1^2 \tau^3 - \beta\tau\right], \quad \alpha_3, \beta \geq 0. \tag{19}$$

For the Brownian preconditioning model, $\alpha_3 = D_v/3$. The parameter β is an effective linear-time attenuation rate, not a named microscopic collision frequency unless a chosen kinetic operator and asymptotic regime justify that mapping. In a dimensionless simulation it is simply a nonnegative coefficient. Figure 3a separates the cubic, linear, and product factors for illustrative values; the curves are formula evaluations, not fits to experimental data.

6.3 Parameter identifiability

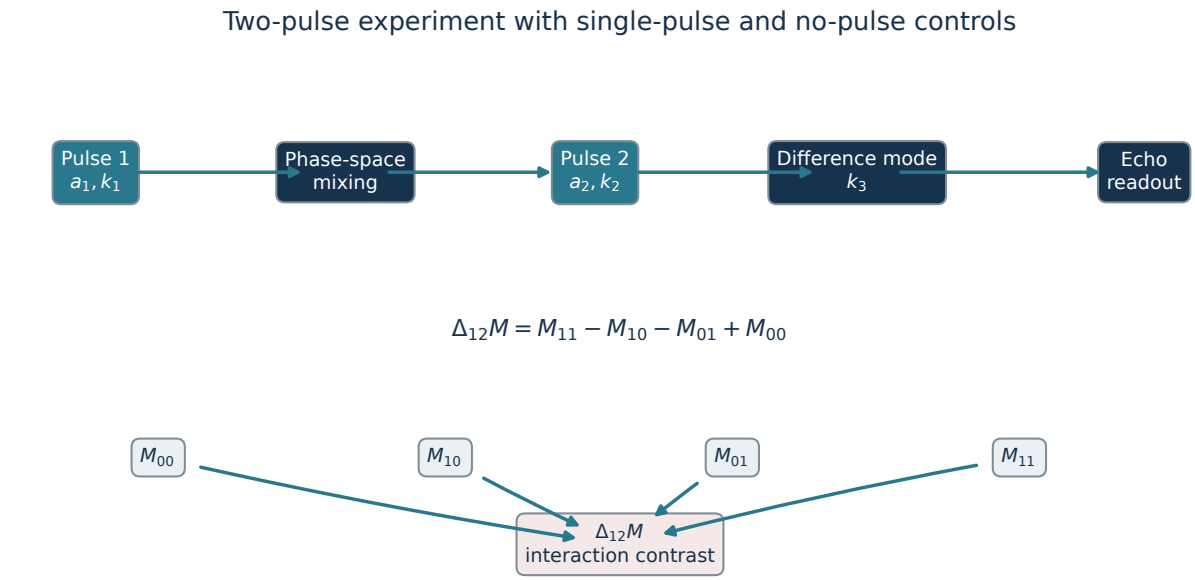
Suppose a pulse-interaction observable has been normalised by a known or independently calibrated collisionless factor G_j and by the pulse product. For positive measured magnitudes, define

$$r_j = -\log\left(\frac{|\Delta_{12}M(\tau_j)|}{|a_1 a_2| G_j}\right). \tag{20}$$

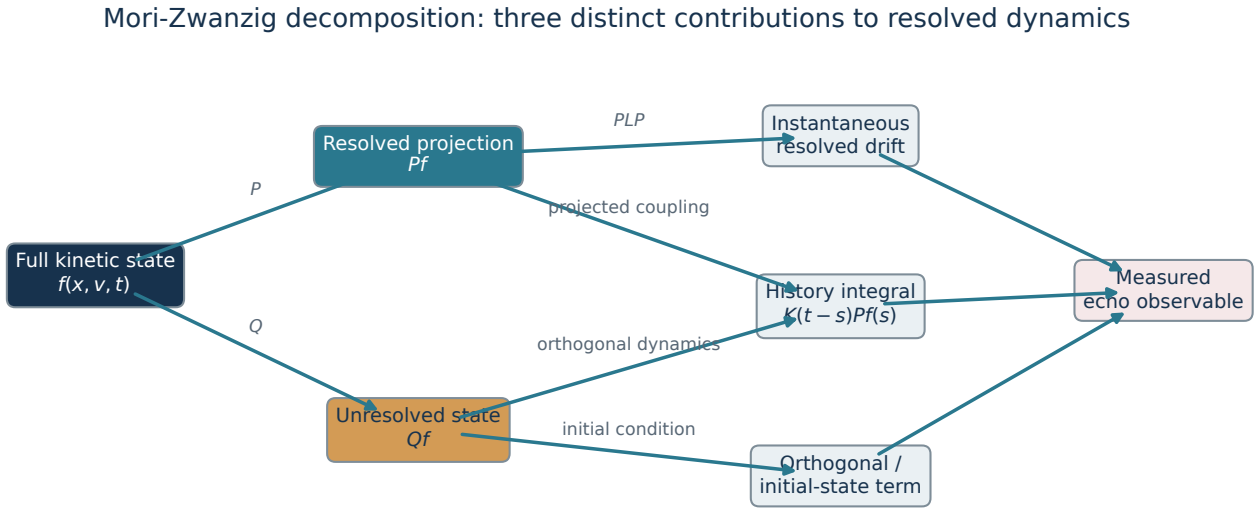
Under Eq. (19), a two-parameter idealisation gives

$$r_j = \alpha_3 k_1^2 \tau_j^3 + \beta\tau_j + \epsilon_j. \tag{21}$$

With k_1 fixed, the design matrix has rows $X_j = (k_1^2\tau_j^3, \tau_j)$. The parameter pair is locally identifiable in the noiseless linear



(a) Two-pulse sequence and four-arm factorial contrast. The interaction term subtracts the no-pulse and single-pulse responses.



(b) Structural decomposition of resolved dynamics into instantaneous, memory, and orthogonal/initial-state contributions.

Figure 2: Two distinct structural views: experimental intervention bookkeeping and projected-dynamics information flow. Both are conceptual diagrams, not empirical causal graphs.

model if and only if $\operatorname{rank}(X) = 2$. For two positive pulse separations τ_1 and τ_2, the determinant of the square two-row design is

$$\det X = k_1^2 \tau_1 \tau_2 (\tau_1^2 - \tau_2^2). \tag{22}$$

It is nonzero exactly when $\tau_1 \neq \tau_2$. Repeated separations cannot separate the cubic and linear coefficients. With more observations, the Fisher-information matrix under Gaussian errors is $\mathcal{I} = X^\top W X$, where W is the inverse noise covariance; full rank is necessary, while a large condition number signals practical instability.

The formal rank condition does not guarantee useful estimation. Large τ increases the contrast between cubic and linear contributions but may attenuate the echo below the measurement floor. Small τ improves signal amplitude but can make the cubic contribution too weak to distinguish. A defensible design therefore optimises a noise-aware criterion such as $\log\det(\mathcal{I} + \lambda I)$ over a bounded admissible range, subject to a minimum detectable echo amplitude and instrument constraints. Multiple k_1 values add a second source of leverage because the diffusion contribution scales as k_1^2 in this reduced family. They also introduce sensitivity to wavevector calibration and mode coupling, which must be included in the uncertainty budget.

7 Connection to Mori–Zwanzig dynamics

Let the full state $z(t)$ obey an autonomous evolution $\dot{z} = R(z)$, and let $\mathcal{L}$ denote the associated Liouville generator. Let P be a projection onto resolved observables and $Q = I - P$. The exact Mori–Zwanzig identity can be written schematically as

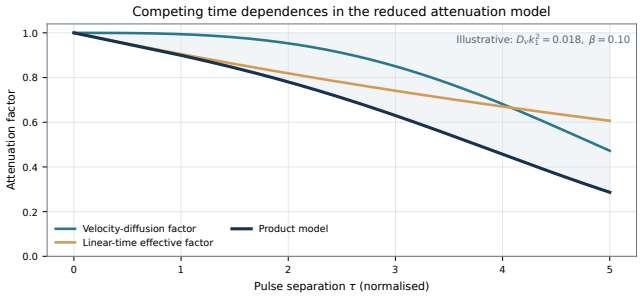
$$\frac{\mathrm{d}}{\mathrm{d}t} P z(t) = P\mathcal{L} P z(t) + P\mathcal{L} \mathrm{e}^{tQ\mathcal{L}} Q z(0) \tag{23}$$

$$+ \int_0^t P\mathcal{L} \mathrm{e}^{(t-s)Q\mathcal{L}} Q\mathcal{L} P z(s)\, \mathrm{d}s. \tag{24}$$

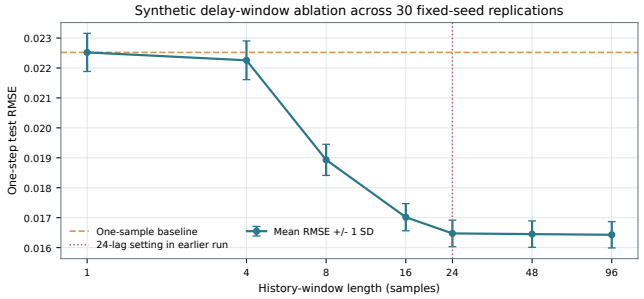
Here A_0 is an observable at the initial time. The identity is the Dyson/Duhamel form of the Mori–Zwanzig decomposition:

Table 4: Attenuation families, interpretable parameters, and identifiability limits. The table is a model comparison, not a claim that these forms exhaust kinetic collision physics.

Family	Normalised form	What it can represent	Limitation / test required
Collisionless phase mixing	$G(t) = \exp[-\sigma_v^2 q(t)^2/2]$	Loss and recovery of a resolved moment through phase cancellation.	Requires a velocity-distribution model; not irreversible field damping.
Cubic diffusion proxy	$\exp[-\alpha_3 k_1^2 \tau^3]$	Integrated Brownian velocity diffusion on a specified interval.	Not a complete Coulomb or Landau collision operator.
Linear-time proxy	$\exp[-\beta\tau]$	Effective exponential loss over the preconditioning interval.	Microscopic interpretation depends on the chosen collision model.
Two-term family	$\exp[-\alpha_3 k_1^2 \tau^3 - \beta\tau]$	Allows both time dependences to be estimated.	Needs distinct pulse gaps, adequate SNR, and operator comparisons.
Empirical operator model	Numerically measured $A_{\mathrm{att}}(\tau, k, C)$	Captures the response of a specified kinetic collision operator C.	Requires converged kinetic simulations, collision-operator specification, and held-out protocols.



(a) Illustrative product attenuation with cubic diffusion and linear-time components. Coefficients are chosen for visual separation, not fitted.



(b) Synthetic one-step RMSE by history-window length across 30 fixed-seed replications. Error bars show one standard deviation.

Figure 3: Two separate model-analysis results: reduced attenuation structure and synthetic temporal-context ablation. Neither panel is empirical plasma evidence.

the first term carries the projected initial tendency under full evolution, the second is the orthogonal-dynamics (often called fluctuating or noise) term, and the last term is the memory contribution. In a chosen resolved basis and inner product, this representation can be reorganised into a generalized Langevin equation with an explicit memory kernel. Equivalent formulations depend on the precise projection and state space. Equation (24) is an identity under the operator assumptions; it does not imply that a finite lag of observations is equal to the exact memory kernel.

An echo experiment can be understood as a controlled perturbation of the unresolved phase-space structure followed by a readout of one resolved Fourier mode. The factorial contrast in Eq. (11) is therefore a candidate input-output probe of how hidden state affects a chosen observable. A reduced model that is Markovian in the chosen resolved state may still fail under pulse histories that share the same current resolved moment but differ in their phase-space filaments. A memory model may improve prediction by retaining information about those histories. However, prediction improvement alone does not identify a unique kernel: alternative hidden-state embeddings, coarse variables, or nonstationary covariates can produce the same forecast gain.

The strongest interpretation is conditional. If two kinetic states have matched resolved variables but controlled differences in their phase-mixed histories, and those differences systematically alter the later echo contrast, then the chosen resolved state is insufficient for that intervention family. That is evidence for missing state information in the reduced representation. It is not, without further assumptions, proof that a particular finite-window convolution is the unique closure. A subsequent study must compare explicit convolutional kernels, delay-coordinate models, latent-state models, and physically derived moment closures on matched train/test partitions.

8 Synthetic forecasting experiment

8.1 Signal and protocol

The software experiment is intentionally separated from the kinetic argument. A scalar deterministic signal is built from a damped oscillation with a slowly modulated phase and a localised Gaussian pulse, then contaminated with Gaussian noise:

$$y(t) = \mathrm{e}^{-0.012t}\sin\big(0.72t + 0.12\sin(0.11t)\big) + 0.38\mathrm{e}^{-(t-55)^2/(2\cdot 1.8^2)} + \eta(t), \tag{25}$$

where $\eta(t) \sim \mathcal{N}(0, 0.015^2)$. The localised pulse is an echo-like synthetic feature only. It is not generated by a kinetic equation and is not labelled as a plasma echo. The sampling interval is 0.05 and the sequence has 2400 samples.

For a history length L, the feature vector contains the current and preceding $L-1$ scalar observations. A ridge-regression predictor with penalty $\lambda = 0.01$ predicts the next sample. The one-sample model ($L = 1$) is a deliberately minimal comparator; it is not equivalent to every possible memoryless closure, because a realistic closure could use multiple resolved moments. The chronological split uses the first 70% of the common sample index set for fitting and the final 30% for testing. To make history lengths comparable, all lags use the

Table 5: Synthetic lag-window ablation over 30 fixed-seed replications. Values are mean RMSE and sample standard deviation across seeds.

Lag L	Mean RMSE	SD
1	0.022520	0.000638
4	0.022257	0.000647
8	0.018932	0.000520
16	0.017017	0.000453
24	0.016475	0.000443
48	0.016452	0.000441
96	0.016429	0.000439

same target indices starting at the maximum history length of 96 samples.

Thirty independent seeds, 20261110 through 20261139, are run. The lag lengths are $L \in \{1, 4, 8, 16, 24, 48, 96\}$. The metric is one-step test root-mean-square error,

$$\mathrm{RMSE} = \sqrt{\frac{1}{N_{\mathrm{test}}} \sum_{n \in \mathrm{test}} (\widehat{y}_{n+1} - y_{n+1})^2}. \tag{26}$$

No hyperparameter search is performed on the test set. The repeated-seed experiment measures sensitivity to the injected noise realisation under this one signal generator; it does not measure generalisation across physical equilibria or kinetic regimes.

8.2 Results

At 24 lags, the mean RMSE is 0.016475 (sample standard deviation 0.000443), versus 0.022520 (0.000638) for the one-sample comparator. The mean per-seed relative improvement is 26.83%, with a two-sided 95% t-interval half-width of 0.51 percentage points; the 24-lag model wins in 30 of 30 replications. In the lag ablation, mean RMSE decreases from 0.022520 at $L = 1$ to 0.018932 at $L = 8$, 0.017017 at $L = 16$, and 0.016475 at $L = 24$, then changes only slightly at $L = 48$ and $L = 96$. The pattern is consistent with the localised deterministic feature being predictable from recent history, but the experiment cannot distinguish physical memory from ordinary autocorrelation and signal-generator structure.

8.3 Interpretation and negative controls

The long-window plateau is important: a lower forecast error is not evidence that a particular 24-step memory horizon is physically correct. To test whether a learned gain arises from meaningful temporal structure, future experiments must include (i) history shuffling within matched state bins, (ii) time reversal or phase randomisation where physically admissible, (iii) a current-state baseline with all resolved moments rather than a single scalar, (iv) lag-window ablation selected without test leakage, and (v) grouped holdouts over pulse timing, wavenumber, amplitude, equilibrium distribution, collision strength, and numerical resolution. If the gain disappears when histories are shuffled while instantaneous observables are preserved, that is evidence that temporal order matters; if it survives, the model may be exploiting static covariates or data leakage. If gain exists only for the generator's fixed pulse shape, it is not a general closure result.

9 Proposed kinetic validation programme

The next scientific gate is a self-consistent Vlasov–Poisson computation. A minimum electrostatic one-dimensional system is

$$\partial_t f + v\partial_x f + \frac{q}{m} E \partial_v f = C[f] + S(x, v, t), \tag{27}$$

$$\partial_x E = \frac{q}{\epsilon_0} \left(\int f \, \mathrm{d}v - n_0 \right), \tag{28}$$

where $C[f]$ is a specified collision operator and S represents controlled pulse injection or a mathematically equivalent initial perturbation. The collisionless baseline must first reproduce a trusted Landau-damping benchmark before any echo inference is attempted. The numerical method must document velocity and spatial domains, truncation, boundary conditions, time integration, field solve, dealiasing or filtering, and resolution-convergence criteria.

A recommended protocol is: (1) initialise an equilibrium that satisfies the stated stability condition; (2) apply a small first perturbation; (3) allow phase mixing over τ; (4) apply the second perturbation; (5) track the mixed mode k_3 over a window centred on the predicted echo time; (6) repeat all four pulse-control arms to obtain $\Delta_{12}M$; and (7) repeat across pulse parameters, collision operators, and grids. The two-dimensional vector result motivates additional oblique or multidimensional tests only after the one-dimensional system is validated. Figure 4 shows the validation and inference stages as two linked but distinct procedures.

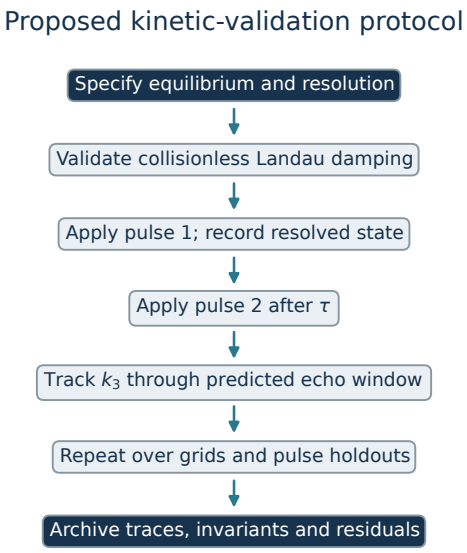
9.1 Numerical acceptance conditions

A future kinetic result should not be interpreted until the following checks pass:

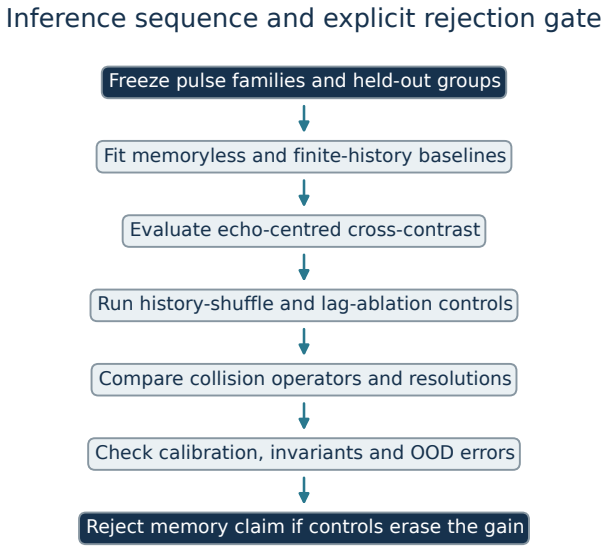
- **Collisionless baseline:** reproduce a trusted linear Landau-damping benchmark and verify the expected dependence on equilibrium parameters.
- **Echo convergence:** predicted and measured echo times converge under simultaneous refinement of Δx, Δv, and Δt.
- **Invariant monitoring:** report particle number, momentum, and energy behaviour appropriate to the collision operator and discretisation; quantify numerical dissipation separately from physical collisional loss.
- **Pulse linearity:** show a small-amplitude range where $\Delta_{12}M$ scales bilinearly and quantify departures from the perturbative regime.
- **Operator dependence:** compare collisionless, velocity-diffusion, and at least one physically motivated collision operator; do not calibrate an empirical β on one operator and transfer it without validation.
- **Out-of-distribution test:** hold out complete pulse protocols and kinetic parameter families, rather than random adjacent time samples.

10 Competing explanations and failure analysis

An echo-amplitude deficit is not uniquely attributable to collisions. Candidate explanations include finite pulse duration, imperfect pulse synchronisation, wavevector calibration error, finite velocity-domain truncation, numerical filtering, field-particle coupling, nonlinear trapping, a non-Maxwellian equilibrium, spatial inhomogeneity, and measurement bandwidth. Each can alter the observed peak, width, or amplitude. The factorial contrast removes additive single-pulse contributions only if the four protocol arms are otherwise matched; it does



(a) Proposed kinetic-solver validation sequence.



(b) Inference and explicit rejection sequence for memory claims.

Figure 4: Two flow diagrams for a future experiment. They are proposed procedures, not completed simulations.

not remove nonlinear cross-talk unrelated to the hypothesised mechanism.

A finite-memory predictor can likewise appear successful for reasons unrelated to physical unresolved dynamics. Leakage across chronological splits, a deterministic pulse template repeated between train and test, a drifting mean, hidden protocol metadata, or a poor baseline can create a spurious apparent gain. For this reason, future comparisons must split by entire protocol and initial-condition family, calibrate uncertainty, report performance across the echo window as well as on average, and include controls that destroy temporal ordering while preserving instantaneous statistics. A model should be rejected when its advantage is not robust to these interventions.

10.1 Candidate scope decisions

A broad search of adjacent directions was screened before fixing the paper's scope. The scores in Table 7 are qualitative internal judgements on a five-point scale; they are not bibliometrics, probability estimates, or claims of objective rank. The chosen scope is intentionally narrower than a new neural closure or collision theory because those would require kinetic training data and operator validation not available in this first release.

11 Limitations

The most important limitation is structural: the present numerical implementation does not solve Eqs. (27)–(28). It has no self-consistent electric field, no nonlinear trapping, no Landau contour calculation, no physical collision operator, and no multidimensional kinetic geometry. The Gaussian velocity distribution is an analytic device that makes the characteristic function explicit. A non-Gaussian equilibrium would have a different characteristic function and potentially different envelope structure.

The Brownian velocity-diffusion calculation is exact for the stochastic process in Eq. (14), over the specified finite interval. It is not a derivation of the entire echo envelope

Table 6: Proposed validation and falsification matrix. Each row names a claim, the control required, and a result that would count against it.

Claim under test	Required test	Supportive outcome	Falsifying or disconfirming outcome
Kinematic echo timing	Sweep k_1, k_2, τ and compare the measured k_3 response with Eq. (3).	Peak time converges to the predicted value as resolution increases.	Persistent timing bias, mode misidentification, or non-convergence.
Bilinear cross-pulse response	Four-arm contrast over a decreasing amplitude ladder.	$\Delta_{12}M/(a_1a_2)$ approaches an amplitude-independent plateau.	No stable plateau or sign/mode changes under refinement.
Diffusive cubic attenuation	Compare collisionless and velocity-diffusion runs across k_1 and τ.	Log attenuation follows the model within a stated asymptotic regime.	Residuals depend strongly on grid or fail the predicted $k_1^2\tau^3$ structure.
Separate α_3 and β	Use distinct pulse gaps with weighted design rank two and adequate echo SNR.	Parameter estimates are stable across design subsets.	Rank deficiency, broad confidence intervals, or parameter trade-off.
Reduced-state insufficiency	Match resolved moments while varying phase-space history; compare echo contrasts.	Echo response depends on controlled history after matching resolved state.	No history effect above uncertainty, or effect vanishes under resolution refinement.
Useful learned memory	Compare matched Markov, delay, and kernel baselines on grouped held-out protocols.	History model improves calibrated prediction under unseen pulse families.	Gain disappears under history controls or is restricted to training protocols.
Physical collision attribution	Compare multiple collision operators and quantify discretisation error.	Parameters map consistently only within a justified operator regime.	Apparent attenuation is explained by numerical dissipation or operator-specific mismatch.

under a Coulomb, Lenard–Bernstein, Dougherty, or Landau collision operator. The added β term is phenomenological and is included to make model uncertainty explicit. Parameter identification requires a calibrated collisionless factor, adequate signal-to-noise, matched controls, and a design matrix with sufficient rank. In realistic data, the pulse amplitudes, transfer function, and noise covariance may also be uncertain.

The vector criterion assumes a Gaussian velocity ensemble and a single scalar observation time applied to the vector phase. Anisotropic or multimodal distributions require their own characteristic functions. The scalar synthetic regression benchmark has a single hand-constructed signal generator, one-step prediction target, fixed ridge regularisation, and a chronological holdout. Its 30 replications vary only the noise seed. The delay-window improvement may arise from autocorrelation and signal-generator structure; no physical memory kernel is identified. Finally, the literature review is targeted rather than a systematic review or patent search, so the absence of a cited method is not evidence of absolute priority.

12 Conclusion

A plasma echo is valuable as a phase-sensitive intervention because it can turn hidden velocity-space structure into a later resolved signal. The scalar rephasing relation follows directly from the stated two-pulse phase ansatz. For a Gaussian velocity ensemble, the echo-mode coherence is a characteristic function whose vector extension makes incomplete rephasing under non-collinear geometry explicit. A four-arm factorial contrast offers a disciplined way to isolate cross-pulse response, while the integrated-Brownian calculation explains the cubic time dependence for one reduced diffusion process. A two-term attenuation model makes the identifiability problem visible: multiple distinct pulse separations and sufficient signal-to-noise are required to separate cubic and linear contributions.

The central scientific caution is equally important. Echo-conditioned response can test whether a selected reduced state is missing history-dependent information, but it cannot by itself recover a unique Mori–Zwanzig kernel. The synthetic delay-window result is a software benchmark, not a plasma result. The next decisive step is a converged self-consistent Vlasov–Poisson study with matched pulse controls, physical collision operators, invariant monitoring, grouped holdouts, and independent numerical convergence. Until those tests are completed, the contribution is a mathematical/reduced-order framework and falsification protocol, not a newly established plasma mechanism.

Reproducibility, data availability, and author responsibility

The analytic calculations and synthetic benchmark are specified in this manuscript. The companion verification script uses Python, NumPy, SciPy, and Matplotlib; the random seeds, sample count, lag set, and ridge penalty are stated in Section 8. The synthetic lag-ablation table is generated from 30 fixed seeds. The plots are generated from the same formulas and code. There are no external kinetic datasets, laboratory records, spacecraft observations, or private data in the numerical experiment. No DOI or external repository deposit is claimed by this document. All numerical outputs are synthetic or reduced-model calculations, and the paper should be cited as a non-peer-reviewed preprint until reviewed through an appropriate scholarly process.

Table 7: Internal topic screening. Scores are qualitative (1–5) and record why the present work is a reduced methodological study rather than a stronger physical claim.

Candidate direction	Gap	Math	AI role	Validation	Falsifiability	Decision / reason
Echo-conditioned memory validation	4	5	4	4	5	Select; a controlled intervention can be formulated now, with a clearly staged kinetic validation.
Neural-operator Landau closure	1	5	5	4	4	Do not claim as new; direct 2026 precedent exists.
Hermite-cascade inversion	3	5	4	2	4	Defer; requires a resolved kinetic solver and phase-space output.
Collision-operator identification	4	4	4	2	5	Defer; requires multiple validated collision operators.
MMS velocity-space echo diagnostics	3	4	3	2	4	Defer; spacecraft data do not provide a controlled two-pulse intervention.
Nonlinear echo-cascade forecasting	4	5	4	2	5	Defer; the scope is broad and demands a kinetic data programme.
Memory-kernel identifiability theorem	3	5	3	2	4	Defer; no new theorem is established in this release.
Coulomb-collision echo digital twin	4	4	4	1	4	Defer; no physical parameter mapping or validated collision solver is available.

Co-equal research-process credit is given to Devanik Debnath and Xylia, the autonomous multi-model research engine coordinating AI-assisted discovery, synthesis, and verification. Devanik Debnath retains responsibility for the final manuscript, scientific claims, citations, and released artifacts.

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A Derivation of the vector optimum

Starting with $J(t) = (\boldsymbol{k}_2\tau - \boldsymbol{k}_3 t)^\top \Sigma_v (\boldsymbol{k}_2\tau - \boldsymbol{k}_3 t)$, expand

$$J(t) = \tau^2 \boldsymbol{k}_2^\top \Sigma_v \boldsymbol{k}_2 - 2t\tau \boldsymbol{k}_3^\top \Sigma_v \boldsymbol{k}_2 + t^2 \boldsymbol{k}_3^\top \Sigma_v \boldsymbol{k}_3. \tag{29}$$

Since Σ_v is positive definite and $\boldsymbol{k}_3 \neq 0$, the coefficient of t^2 is positive. Differentiation gives Eq. (10). Substitution gives the unconstrained minimum

$$J_{\min} = \tau^2 \left[\boldsymbol{k}_2^\top \Sigma_v \boldsymbol{k}_2 - \frac{(\boldsymbol{k}_3^\top \Sigma_v \boldsymbol{k}_2)^2}{\boldsymbol{k}_3^\top \Sigma_v \boldsymbol{k}_3} \right] \geq 0, \tag{30}$$

where non-negativity follows from Cauchy–Schwarz in the inner product induced by Σ_v. The unconstrained maximum coherence is $\exp(-J_{\min}/2)$. The post-pulse restriction $t \geq \tau$ replaces the unconstrained optimum by the nearest admissible point because $J(t)$ is convex.

B Identifiability and design conditioning

For m observations with distinct positive pulse separations, let $X \in \mathbb{R}^{m\times 2}$ have columns $k_1^2\tau_j^3$ and τ_j. If the weights are positive, $X^\top W X$ is positive definite exactly when X has full column rank. The determinant condition in Eq. (22) is therefore necessary and sufficient for two distinct nonzero pulse gaps in the two-observation case. More observations do not automatically improve conditioning: if the selected τ_j values occupy a narrow range, the two columns can remain nearly collinear. Very wide ranges improve geometric separation but may move long-gap echoes below the detection threshold. The design problem is consequently constrained and noise-dependent; a rank test is the first gate, not the final criterion.

C Synthetic benchmark details

For each seed, $n = 2400$ observations are generated at $\Delta t = 0.05$. The common target index begins at 96 so that all lag lengths share identical targets. A chronological 70/30 split is applied to the resulting one-step prediction pairs. For each lag L, the regressor contains $y_{i-L+1}, \ldots, y_i$ and the response is y_{i+1}. Ridge coefficients solve $(X^\top X + \lambda I)\hat{\theta} = X^\top y$ after adding an unpenalised intercept, with $\lambda = 0.01$. The reported standard deviations are sample standard deviations across the 30 seed-level RMSE values. The 95% interval uses a Student-t critical value on the 30 per-seed relative improvements. These intervals quantify seed variation under the specified generator only; they do not express uncertainty across physical regimes.

D Finite-sample uncertainty and detection-limited design

The factorial contrast is a constructed estimator and its uncertainty must be propagated from the four experimental arms. Write

$$\widehat{\Delta}_{12} = \boldsymbol{c}^\top \boldsymbol{M}, \qquad \boldsymbol{c} = (1, -1, -1, 1)^\top, \tag{31}$$

where $\boldsymbol{M} = (M_{11}, M_{10}, M_{01}, M_{00})^\top$ follows the ordering in Eq. (11). If the arm-wise measurement errors have covariance matrix Σ_M, then

$$\mathrm{Var}(\widehat{\Delta}_{12}) = \boldsymbol{c}^\top \Sigma_M \boldsymbol{c}. \tag{32}$$

For independent, equally noisy arms with variance σ^2, Eq. (32) reduces to $4\sigma^2$. That special case should not be assumed when simulations use common initial conditions, paired instrument records, or common random numbers. Positive cross-arm covariance can reduce the variance of the contrast because the signs in $\boldsymbol{c}$ cancel shared fluctuations; negative covariance can increase it. The covariance should therefore be estimated from replicated protocol blocks, not inferred from a single time trace. Pairing the four arms by equilibrium realisation and instrument configuration is desirable, but only if the pairing preserves the physical comparability of the interventions.

The log-linear attenuation statistic in Section 6 also changes the noise model. Let $\widehat{r} = -\log(|\widehat{\Delta}|/(|a_1 a_2|G))$, where G is a calibrated collisionless response factor. Away from zero and for relative errors small compared with the contrast magnitude, a first-order delta-method approximation gives

$$\mathrm{Var}(\widehat{r}) \simeq \frac{\mathrm{Var}(\widehat{\Delta})}{\Delta^2} + \mathrm{Var}(\log G) + \mathrm{Var}(\log |a_1 a_2|), \tag{33}$$

when the three calibration sources are mutually independent. Covariance terms must be added otherwise. The leading term grows rapidly as $|\Delta|$ approaches the noise floor. Thus an unweighted least-squares fit to $\widehat{r}$ is generally inefficient and may be seriously biased when low-amplitude echoes are retained or discarded by a threshold. The approximate log-variance is useful for design and initial weighting, but it is not a substitute for a likelihood model when the signal-to-noise ratio is small.

For observations that remain comfortably above the detection threshold, define $\boldsymbol{r} = (\widehat{r}_1, \ldots, \widehat{r}_m)^\top$, $X_{j,:} = (k_{1,j}^2 \tau_j^3, \tau_j)$, and a covariance matrix Σ_r. Generalised least squares gives

$$\widehat{\boldsymbol{\theta}} = (X^\top \Sigma_r^{-1} X)^{-1} X^\top \Sigma_r^{-1} \boldsymbol{r}, \qquad \boldsymbol{\theta} = (\alpha_3, \beta)^\top. \tag{34}$$

If Σ_r is known and the linearised model is correct, the corresponding covariance is $(X^\top \Sigma_r^{-1} X)^{-1}$. In practice, residual model discrepancy, uncertainty in G, pulse-amplitude calibration, and run-to-run variability must also enter the interval estimate. Profile likelihood or a block bootstrap over complete equilibrium/protocol replicates is preferable to resampling adjacent time samples, which are temporally dependent. The confidence region should be reported jointly for (α_3, β); separate marginal intervals can conceal a strong trade-off between the two coefficients.

A second constraint is detectability. If $\sigma_\Delta(\tau)$ denotes the standard deviation of the factorial contrast and q is a predeclared detection multiplier, a protocol is retained only when its predicted interaction magnitude satisfies

$$|a_1 a_2| G(\tau) \exp[-\alpha_3 k_1^2 \tau^3 - \beta\tau] \geq q\, \sigma_\Delta(\tau). \tag{35}$$

Table 8: Noise-aware design diagnostics. The entries are methodological requirements, not results from a completed plasma experiment.

Failure mode	Required diagnostic or remedy
Unequal arm noise	Estimate the full four-arm covariance and propagate it through the signed contrast.
Shared run fluctuations	Pair arms by equilibrium block and retain cross-arm covariance.
Small echo magnitude	Use a heteroscedastic or censored likelihood; do not silently drop non-detections.
Calibration uncertainty	Propagate uncertainty in G, a_1, and a_2 into the attenuation fit.
Collinear design columns	Report the weighted condition number and joint confidence region for (α_3, β).
Temporal dependence	Bootstrap or split complete protocol/equilibrium blocks, not adjacent samples.
Mode contamination	Retain complex Fourier coefficients and inspect phase and neighbouring-mode residuals.

Equation (35) defines a practical, noise-dependent upper range for τ; it is not a physical boundary. The multiplier q must be fixed before inspecting the response, with its false-positive interpretation stated for the chosen noise distribution and multiplicity of tested modes. Optimising only the determinant of the information matrix can select pulse gaps whose echoes are mathematically informative but experimentally unobservable. A defensible design instead maximises a criterion such as $\log\det(X^\top \Sigma_r^{-1} X)$ subject to Eq. (35), pulse linearity, and the available observation window.

Below the detection floor, taking the logarithm of a noisy magnitude and deleting non-detections is not neutral: it preferentially removes the most attenuated realisations and can bias the inferred coefficients downward. Such measurements should be handled as censored observations or analysed in the original contrast domain with an explicit noise model. If the complex Fourier coefficient is available, retaining its real and imaginary components is preferable to reducing it immediately to magnitude, because phase mismatch and mode contamination can then be diagnosed separately. A robust analysis should report the detection fraction by pulse family, the contrast covariance, the complex-mode residuals, and the sensitivity of $(\widehat{\alpha}_3, \widehat{\beta})$ to the detection rule.

These conditions make the identification claim deliberately conditional. A full-rank design is necessary for estimating two attenuation coefficients, but it is not sufficient when the echo is undetectable, the collisionless calibration is uncertain, or the assumed response family is misspecified. The proposed protocol should therefore report both algebraic identifiability and practical identifiability under an explicit noise and detection model. A parameter estimate without those diagnostics is not evidence that the underlying microscopic process has been identified.